

Tejal Anavekar

CS graduate from USC with experience at Meta developing large-scale data pipelines and analytics platforms, complemented by full-stack engineering, data infrastructure, and AI/LLM-powered systems.

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EDUCATION

University of Southern California

Los Angeles, CA

Master of Science in Computer Science | Graduated May 2026

- Relevant coursework: Analysis of Algorithms, Database management, Applied Natural Language Processing, Information Retrieval, Geospatial Information Management

Fr.C.Rodrigues Institute of Technology

Mumbai, India

Bachelor of Engineering in Information Technology | Graduated June 2024

- Relevant coursework: Operating Systems, Computer Networks, Artificial Intelligence, Big Data, Cloud Computing

TECHNICAL SKILLS

Languages: Python, C, C#, HTML, CSS, SQL, Javascript, Typescript

Engineering Practices: Data Pipelines, REST API Design, System Design, OOP, ETL/ELT

Frameworks: React, Node.js, Express.js, ASP.NET Core, REST APIs, Minimal API, Flask

Libraries & AI Frameworks: LangChain, AI Agents, PyTorch, TensorFlow, Scikit-learn, OpenCV, Pandas, NumPy

Databases: PostgreSQL, MongoDB, Supabase, Firebase, SQL Server (T-SQL), Redis

Cloud & Tools: AWS (EC2, S3), GCP, Git, Docker, Linux/Unix, Shell Scripting

EXPERIENCE

Data Engineer Intern

June 2025 – Aug 2025

Meta Platforms

Burlingame, CA

- Built and productionized 4 large-scale data pipelines on Meta's internal infrastructure, processing billions of records daily through distributed ingestion to drive real-time analytics and AI readiness dashboards across internal platform operations
- Developed Python-based data validation frameworks enforcing schema integrity, null-safety, and idempotency guarantees
- Optimized DAG-based data workflows through parallel execution with solutions adopted by 3+ cross-functional teams

Software Developer

Sept 2024 – May 2026

USC Institute for Creative Technologies

Los Angeles, CA

- Engineered full-stack internal platforms using C# ASP.NET Core Razor Pages, integrating Minimal API endpoints and dependency injection patterns for efficient and structured payload delivery across internal services
- Architected normalized SQL Server schemas, stored procedures, and EF Core data access layers, leveraging LINQ query optimization to accelerate dynamic reporting workflows and reduce Excel report generation time by 40%
- Deployed real-time dashboards to production, implementing end-to-end data validation and monitoring at live environment

AI Engineer

June 2022 – Apr 2023

Tata Institute of Fundamental Research

Mumbai, India

- Built CV pipelines integrating SIFT feature extraction, improving classification precision by 25% over CNN baselines
- Fine-tuned and deployed ResNet-50 and VGG variants, reducing inference latency by 30% in high-throughput workloads
- Scaled modular preprocessing pipelines like resolution normalization, histogram equalization, brightness calibration for robust, standardized input handling across heterogeneous datasets in production-grade ML pipelines

PROJECTS

FixIt - AI Powered Debugging Platform | React, Typescript, Node, Express, Supabase, Groq LLM

June 2026 – Present

- Architected a debugging platform transforming software errors into interactive learning workflows and adaptive assessments
- Engineered schema-constrained LLM pipelines with Zod validation across 12+ structured fields, transforming natural-language debugging queries into syntax breakdowns, concept explanations, and knowledge-validation quizzes
- Implemented session-aware user state management and persistent sandbox execution using Supabase Auth and PostgreSQL

Budget Buddy | MongoDB, Express, React, Node, LangChain, RAG, REST API

Jan 2026 – Present

- Architected a modular full stack system with decoupled service and UI layers, improving scalability and maintainability
- Implemented secure identity and access management workflows using JWT, bcrypt for protected financial records
- Built RAG-powered LangChain pipelines for personalized financial intelligence, improving recommendation accuracy by 25%

Gene Semantic Bridge | Pytorch, CVAE, BioBert, scRNA-seq, Generative Modeling

Jan 2026 – May 2026

- Designed KA-CVAE (Conditional VAE) integrating BioBERT embeddings to model unseen CRISPR perturbations
- Engineered parameterized architecture with 128-dim latent space modeling across single and combinatorial perturbations
- Evaluated on Norman scRNA-seq OOD benchmarks, achieving R-squared value of 0.996, outperforming baseline models

GeoForecast - Weather App | MongoDB, Express, React, Node, Flask, Swift, GCP

Aug 2024 – Dec 2024

- Designed microservices-based system decoupling geolocation, weather, and user services into deployable modules
- Deployed containerized APIs on GCP Cloud with autoscaling, achieving 93% location accuracy, and +35% user retention
- Integrated geolocation and weather REST APIs with caching for low-latency real-time and daily forecast delivery